

## Part B. Documentation and Analysis of Prior Solution Attempts

This study analyzed results of different types of water filters and what they did as far as filtering out the dirtiness of the water and then getting rid of the contaminants and other things in the water.

### **What worked:**

- Water filters that use activated carbon can filter out contaminants quite well.
- Filters that use sand, gravel, and charcoal can filter out the dirtiness but can't filter the contaminants.
- Filters that use many different stages and let the water go through different minerals, filters, and other things.

### **What didn't work:**

- Filters that don't have the correct materials to remove contaminants and have to let water flow through multiple times to become clear.
- Filters that have the materials in the wrong order.
- Filters that didn't have a way to remove the contaminants out of the water after so it's unsafe to drink.

### **Ways to improve:**

To improve it would be best to run a lot of trial and error experiments trying different things until you find something that works. You could also try putting different materials inside of the filter. In some cases you could even change the design of the filter itself to see if it results in any change. Eventually after many attempts you would finally get one that worked as wanted to use. The one you made that worked could be used to get clean water that is safe to drink available in places that don't have it.